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No. of Printed Pages: 02

SEAT No.________

[12]

SARDAR PATEL UNIVERSITY
External Examinations
M.Sc. (Information Technology) Semester – I
PS01CINT35: Operating Systems
Friday, 26th February, 2021

Max Marks: 70

Time: 10:00 A.M. to 12:00 NOON

Q.1 **[08]**

[A] Choose the most appropriate option for each question.

1 In ___ systems, the CPU executes multiple jobs by switching among them.
- (A) time sharing
- (B) multiprogramming
- (C) multitasking
- (D) Both (A) and (C)

2 __ provide an interface to the services made available by an operating system.
- (A) Function call
- (B) System call
- (C) API
- (D) All of these

3 In the dining philosopher problem, philosophers represent __
- (A) processes
- (B) resources
- (C) waiting queue
- (D) None of these

4 PCB stands for ____________________
- (A) Process Control Box
- (B) Program Control Box
- (C) Process Control Block
- (D) Program Control Block

5 Which of the following is **NOT** relevant in case of memory?
- (A) Space
- (B) Address
- (C) Time
- (D) None of these

6 Which of the following is NOT relevant in case of Paging scheme?
- (A) Frame
- (B) Page
- (C) Page table
- (D) None of these

7 Which of the following algorithm is solution to the critical section problem?
- (A) FIFO
- (B) LRU
- (C) SJF
- (D) None of these

8 Which of the following term is related to total number of bytes transferred?
- (A) Rotational Latency
- (B) Throughput
- (C) Seek Time
- (D) All of these

Q.1 **[16]**

[B] Do as directed/ Fill in the blanks

i. Trap is a software generated interrupt. **[True/False]**

ii. Cache is the fastest memory in terms of access. **[True/False]**

iii. Operating system that provides GUI, can't provide command line interface. **[True/False]**

iv. If a process calls a system call, interrupt occurs. **[True/False]**

v. Signal operation in semaphore decrements the value of variable. **[True/False]**

vi. A process under execution is said to be in ready state. **[True/False]**

vii. The number of processes that are completed per time unit, called ____.

viii. Peterson's solution for critical section problem can be used for only two processes. **[True/False]**

ix What is base register?

x An address generated by the CPU is called ________ address.

xi Paging is a contiguous memory allocation scheme. **[True/False]**

xii Paging can suffer from internal fragmentation. **[True/False]**

xiii ______________ is a named collection of related information that is recorded on secondary storage.

xiv SSTF stands for ____________

xv RAID level 0 provides only mirroring. **[True/False]**

xvi Protection Attribute of a file provides Access-control information determines who can do reading, writing, executing, and so on.**[True/False]**

Q.2 Answer the following questions (Any Seven): [14]

a. Differentiate between the user mode and kernel mode

b. What is command interpreter?

c. List necessary conditions for deadlock.

d. List out various multithreading model.

e. Explain swapping in brief?

f. What is demand paging?

g. List out attributes of files.

h. What are the two main purposes of using RAID?

i. Draw a two level directory structure.

Q3. What is operating system? Explain various services provided by operating systems to users as well as to the machine. [8]

OR

Q.3 Explain following structures of operating systems: [8]
(i) Microkernel (ii) Layered Structure

Q.4 Differentiate between the terms process and program. Draw process state diagram and explain it in detail. [8]

OR

Q.4 Execute FCFS CPU scheduling algorithm for arrival sequence of processes (P1, P2, P3), whose CPU burst time is (24, 10, 2) respectively (in milliseconds). Draw the Gantt Chart and calculate average waiting time. What is convoy effect in case of FCFS algorithm? [8]

Q.5 What do you mean by address binding? Explain different possible steps of address binding [8]

OR

Q.5 Execute optimal page replacement algorithm for following reference string: [8]
3, 0, 2, 4, 3, 2, 6, 7, 1, 8, 7, 0
Calculate total number of page faults for the same. Consider number of frames given are 3.

Q.6 What is file? List different types of files, briefly explain the operations that can be performed on files. [8]

OR

Q.6 List out various algorithms of disk scheduling and explain any one with example. [8]

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